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Scenario analysis of cooking fuel choices and CO₂ emission reduction for bioenergy assessment in rural South-East Nigeria

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Abstract

Rural households in South-Eastern Nigeria rely heavily on traditional biomass fuels for cooking, contributing significantly to carbon dioxide (CO₂) emissions and raising concerns about environmental sustainability and public health. This study conducts a scenario-based analysis to quantify CO₂ emission abatement potentials associated with shifts in cooking fuel types and to assess the viability of bioenergy integration within the regional energy mix. The study employs the Low Emissions Analysis Platform (LEAP) model to conduct a comprehensive scenario analysis of cooking energy demand and CO₂ emission abatement potentials through transitions in SE rural household cooking fuel types. Using region-specific household energy consumption data, emission factors, and techno-economic parameters, three scenarios are simulated over a 37-year horizon: i) Baseline (BLS) with continued traditional biomass use; Short-term (STS); and Long-term scenario (LTS) featuring widespread substitution with modern fuels such as liquefied petroleum gas (LPG) and biogas derived from locally available agricultural residues. Scenario simulations reveal that transitioning to modern bioenergy systems and cleaner fuels can achieve up to a 50% reduction in cooking energy demand due to significant energy efficiency gains, cutting an average of about 650 K GJ of energy demand by 2060. Results showed an unusual paradox that an overdependence on biogas (especially in the long term) could yield more emissions compared to other traditional biomass. Thus, the study revealed a proportional mix of cooking fuel is essential to curbing emissions in rural SE of Nigeria to abate about 350 k MtCO₂e by 2060. The findings provide critical policy recommendations.

Keywords Southeast Nigeria, Bioenergy, Scenario analysis, CO₂ emissions, Rural area, Cooking energy demand, Energy efficiency, LEAP

1 Introduction

Global warming is a pressing issue that has threatened human well-being and the health of the planet and is the paramount cause of human-induced climate change [15]. Moreover, the main agent causing global warming that has exacerbated climate change is



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anthropogenic emissions, especially carbon dioxide (CO₂) [13]. Several international and national efforts and treaties have endeavoured to abate anthropogenic emissions; however, the climate change challenges loom [41]. Among the efforts is abating emissions from the energy sector, which is the largest contributor to global emissions, such as the transportation, agriculture, services, residential and commercial sectors. Although generating electricity and heat by burning fossil fuels such as coal, oil and natural gas is considered one of the largest contributors to global climate change, accounting for a significant portion of greenhouse gas emissions worldwide due to the carbon dioxide released during power production.

However, renewable sources are both sustainable and reliable sources of electricity generation. The process of generating electricity from renewable energy sources, especially solar, wind, geothermal, biowaste and water, is a clean alternative source with diverse benefits on our planet, providing modern energy to underserved regions, especially in developing countries. RE sources like solar and wind have experienced tremendous adoption in many developing countries and have presented easy-to-use and advanced technology. According to recent data, the levelized cost of electricity (LCOE) from solar energy is indeed significantly lower than that of fossil fuel sources, with estimates suggesting it can be around 45% cheaper, making solar a very cost-competitive option for electricity generation [16]. According to a recent report from the International Renewable Energy Agency (IRENA), the global weighted average LCOE of utility-scale solar PV plants declined by 90% between 2010 and 2023, from USD 0.460/kilowatt hour (kWh) to USD 0.044/kWh.

In 2023, the year-on-year LCOE reduction was 12%, 7% and 3% for solar, offshore and on-shore wind, respectively. Between 2010 and 2023, bioenergy experienced around USD 0.084/kWh to USD 0.072/kWh. Also, in 2023, the weighted average LCOE of bioenergy ranged from a low of USD 0.063/kWh in India and USD 0.066/kWh in China to highs of USD 0.097/kWh in Europe and USD 0.107/kWh in North America [16]. It is evidence that Sub-Saharan African countries are not captured in the report, this implies that regions do not deploy bioenergy at a large scale or utilize more traditional biomass to meet their cooking and heating demand, without transformation to provide modern energy [19].

Further, a recent review study by [29]. The study revealed that bioenergy and biogas have not been properly legislated in the Nigerian energy policy; as such, the utilization and adoption of bioenergy and biogas in Nigeria have not received the needed attention, both among the legislators and consumers. Alternatively, a few SSA countries, such as Ethiopia, Tanzania, South Africa, and Kenya, have actively adopted biofuel production using crop residues and other biomass sources, particularly in rural areas [6]. However, challenges remain regarding inefficient usage, environmental concerns, and the need for improved technology and policy frameworks to fully realize its potential.

Nigeria has abundant biomass and produces an estimated 144 million tons of biomass annually, including about 47% agricultural waste, 20% forest (charcoal, 2%; forest residue, 2%; wood fuel, 16%) and 28% animal waste. Thus, about 70% of the people use traditional biomass for cooking and heating [34]. Also, biomass is the major source of energy in the Southeast (SE), Nigeria, used to meet the SE region's cooking and demand needs [8], because the SE region has significant agricultural residues, forest residues and municipal solid wastes [18]. However, the SE region is making significant strides by deploying

bioenergy and converting traditional biomass into a more efficient energy source - biogas and biofuels [17] without a corresponding strategic planning effort on how to adopt and utilize sustainable cooking.

More so, the cooking sector of the region is heavily dependent on wood fire, which has significant environmental and health challenges. Obviously, the SE region is rated low in clean cook adoption, with unsustainable energy practices. The energy challenge has negatively impacted the health of many vulnerable persons and has further impoverished them. As the natural ecosystem and its resources have been largely mismanaged, especially forest resources, such as trees and other organic materials. Thus, making the region susceptible to climate stress. Several studies have identified the challenges of clean cooking adoption in the region, while they have addressed pertinent issues in the region, they have yet to critically investigate CO₂ emission abatement and cooking fuel types in the SE. It is therefore imperative to investigate a framework for the potential assessment of bioenergy adoption and utilization in Nigeria. Thus, exploring a bioenergy framework for regions with a data dearth and limited large-scale adoption could be enhanced by using scenario analysis to prepare for uncertainties and create strategies to achieve desired results in clean cooking [5]. More so, such scenario analysis could enhance improved decision-making by considering multiple possible outcomes, understanding the impact of variables on a situation, identifying potential threats and opportunities and enhancing strategic thinking. Also, scenario analysis is widely deployed to explore carbon dioxide abatement [21]. On this note, the current study aims to use scenario analysis of CO₂ emission and cooking fuel types to explore bioenergy assessment in Southeast Nigeria. Specifically, the study employs the Low Emissions Analysis Platform (LEAP) framework for bioenergy policy analysis. The study contributes significantly to regional energy system modelling, especially towards sustainable cooking, planning rural clean cooking, emissions modelling, and the application of LEAP in data-scarce environments in South-East Nigeria. The study's novelty is attributed to energy modelling methodology and research paradigm (such as emissions reduction, clean cooking and regional study) in areas with significant data dearth, such as rural areas of developing countries.

2 Methodology

2.1 Study area

The study area of the study is Southeast Nigeria. Geographically, Southeast Nigeria is one of the six geopolitical zones in this country. The zone is made up of five States: Abia, Anambra, Ebonyi, Enugu and Imo. Figure 1 shows the five states in the zone and their major cities/capital.

Southeast Nigeria covers an area of about 76,358 km² east of the lower Niger and south of the Benue Valley. The region is located between latitudes 4 and 7° North of the Equator and between longitudes 7 and 9° East [31, 33]. The population of the South East region of Nigeria is around 22 million people, which is about 18% of the country's total population [25]. About 70% of the population of the region lives in rural areas [31].

The geography and climate of the Southeast region are characterized by rainforest, savannah and hilly terrain. According to the Köppen-Geiger climate classification, the SE region belongs to the hot humid climatic zone [36]. The tropical climate has two main seasons, the rainy season from April to October and the dry season from November to

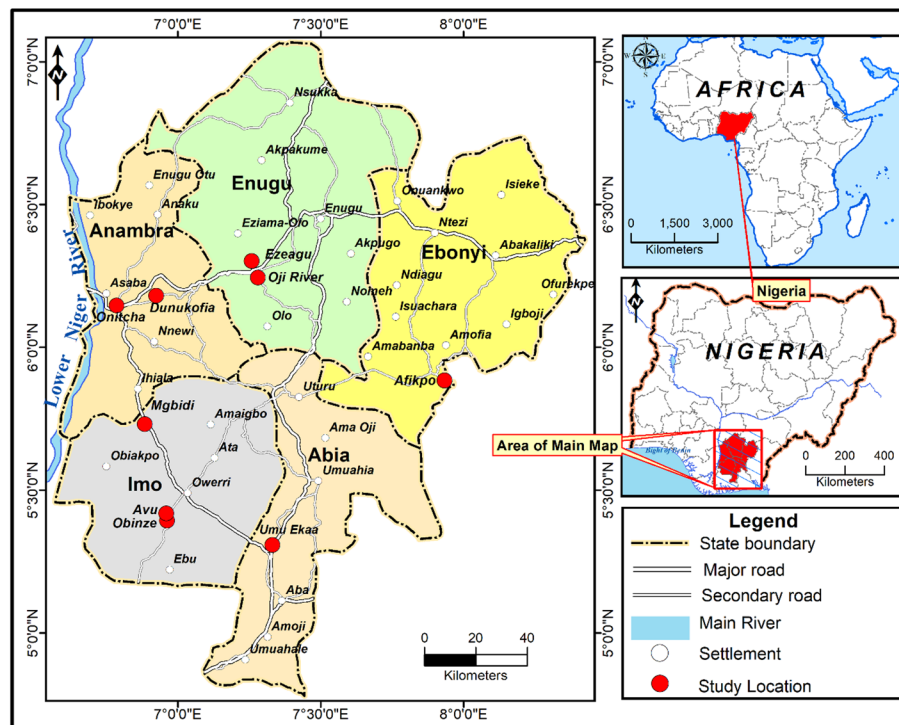


Fig. 1 Map of the South East Region showing Study Locations

March [24]. The Niger River is a major river in the region which borders the west of Anambra State. Southeastern coastal Nigeria consists of low sedimentary plains that are essentially an extension of the southwestern coastal plains [28].

According to the Federal Department of Forestry, the Southeast region is composed of an alluvial soil and a red-yellow ferrallitic soil of humid tropical zones. The zone of alluvial soils contains soils which are formed by recent water deposits and therefore are found on the flooded plains of rivers, or deltas, or along the coastal flats. The soil in the zone is rich in humus and is used to produce cash crops such as cocoa, palm produce and rubber [11]. Also, the region belongs to the derived savanna ecological zone and is regarded as the zone where man's impact on the forests has been high and resulting in a considerable reduction of vegetation/plant species. Thus, remnants of the high forest may be present in upland or rocky areas that are not suitable for agriculture, as such characterized by dense populations [11].

In addition, in terms of land cover characteristics, the region is mainly composed of arable land, traces of grassland land and forest plantations. The arable land cover represents all lands for arable agriculture and the dominant crops cultivated in the region, considering the land type, are yams, cocoyam, cassava, and intermixed with maize. Minor commercial logging activities take place in the region due to the lack of forest plantations [11].

2.2 LEAP scenario model

The Low Emissions Analysis Platform (LEAP) is a powerful and versatile software system for integrated energy, climate change mitigation and air pollution abatement and helps countries create low-carbon energy plans (Stockholm Environment Institute [SEI] [44]. It was developed by the Stockholm Environment Institute to develop projected energy

demand and environmental impact scenarios [9]. It is also a widely used software tool for energy policy analysis and climate change mitigation assessment based on how energy is consumed, transformed and generated in a given region or economy (Stockholm Environment Institute [SEI] [43]. The LEAP model has long helped countries tackle air pollution, health and ecosystem impacts, providing a user-friendly and highly flexible system that boasts thousands of users worldwide and has become crucial to planning under the Paris Agreement. Also, the tool helps countries develop their nationally determined contributions (NDCs), meet Sustainable Development Goals (SDGs) and quantify the benefits of low-emission development pathways (Stockholm Environment Institute [SEI] [44]. It has been extensively adopted in many organisations on the local, national, and international levels to project energy supply and demand, predict the environmental impact of energy policies, and identify possible challenges in the future.

The LEAP model has been used to investigate the scenario analysis of the energy consumption, greenhouse gas emissions across several sectors of the energy system, such as transportation, heating and cooking, also in the commercial, residential, industrial and service sectors. Studies such as [23] investigate through a scenario evaluation using the LEAP model, the prediction of energy consumption and pollutant emissions in Iran, while [1] predicted coal consumption, greenhouse gas (GHGs) emissions in Pakistan. Similarly, [20] investigated the trajectory of carbon emissions from tourist activities in China. Also, [47] analyzed the carbon dioxide (CO₂) emissions and peaks from 2019 to 2035 in Shanxi Province under different scenarios. In a different study, [22] examined how new transport policies could reduce transport energy demand in Lagos State, Nigeria. The study further projected future energy demand and greenhouse gas emissions to determine the most effective solution to achieving a sustainable transport system in the city.

On the application of the LEAP model on bioenergy, [2] investigated the electric power consumption and estimated the energy demand from a landfill gas in the Karachi metropolitan, Pakistan. Similarly, [10] evaluated the biomass energy potential in North Sumatra and analyzed various scenarios of biomass energy utilization in the region. Also, [35] assessed the long-term greenhouse gas mitigation and economic impacts of integrating bioenergy from cassava residues into the South African energy mix. [40] investigated the significance of incorporating biomass feedstocks into Pakistan's energy mix for a sustainable energy future. Additionally, a scientific report through SEI investigated the bioeconomy pathways for different demand scenarios of the usage of bioresources in Rwanda [4]. More recently, [37] The paper forecasts electricity demand for 2030 and develops several electricity scenarios.

In the Nigerian case, [7] analysed the effect of the National Renewable Energy and Energy Efficiency Policy (NREEEP) on final energy demand and CO₂ emissions. Also, several studies have investigated using the LEAP model for the analysis of renewable energy systems, energy sustainability, energy demand-side modelling for the household sector, carbon-neutral economy and transition pathways to carbon neutrality, energy transition to Liquefied Petroleum Gas, low carbon emissions on future oil production [3, 30, 32, 39, 45, 46] accordingly.

Several studies in the literature have focused on the mitigation of GHG emissions, energy demand analysis and renewable energy integration. The studies were conducted by estimating CO₂ emission using the LEAP modelling framework that integrates several

sectors and developing a scenario-based modelling paradigm that can be used to track energy consumption, production and resource extraction in all sectors of an economy by enabling the potentials of diversification of the energy supply mix for national and regional/municipal development.

This study falls into the category where it intends to estimate through CO₂ emission and demand analysis by imposing useful energy demand, energy intensities of technology, efficiency of fuel share and emission factor of the fuel in a bottom-up approach manner through the use of the LEAP model. Thus, no other study has applied a bottom-up approach to estimate an integrated CO₂ emission, demand analysis based on fuel type and corresponding emissions from energy consumption and heating and cooking technologies in the Southeast of Nigeria. Thus, this study intends to fill the gap in the literature.

2.3 The algorithm of the LEAP model

The LEAP model uses a framework for calculating energy consumption, useful energy demand by fuel and carbon emissions analysis. The values are imputed into the LEAP model based on the required data from the five eastern states. These are presented in the following sections.

2.3.1 Demand analysis on energy consumption

The analysis of energy demand and useful energy consumption is based on either the final or useful energy demand analysis. The computation for final energy demand is represented in Eq. 1

$$E_F = \sum_{i=1}^n AL_i EI_i$$

where E_F is the final energy demand; AL_i represents the activity level and is a measure of the social or economic activity for which energy is consumed, and it is measured for each year as the scale of units per household. For instance, to measure the activity level of households, the expression was in the growth (%), the scale was provided in millions of people, and the units were based on per household. The EI_i and the energy intensities of the final consumption of energy per unit of activity level of the respective i th technology branch (such as kerosene, fuelwood, LPG, etc.). It is measured in Gijajoule GJ per metric tonne each year in the period under observation of n , while i is the technology or fuel type.

For each technology branch, the useful energy intensity (U_E) is represented according to Eq. 2

$$U_E = EI_i FS_i \eta_i$$

where FS_i and η_i represents the fuel share and the efficiency, respectively of the i -th technology.

2.3.2 Emission analysis

The greenhouse gas (GHG) emissions are computed in LEAP according to Eq. 3

$$G = \sum_{i=1}^n E \times Ef$$

Where G represents the total GHG emission, E is the energy demand by a given fuel and E_{fj} is the carbon emission factor from fuel type n for equipment j .

2.4 Scenario development and assumptions

The scenarios were formulated based on the National Clean Cooking Policy and the Nigerian Nationally Determined Contributions (Nationally Determined Contribution [NDC] 26; Federal Ministry of Environment [FME] 12. These national documents present clean cooking policies that drive clean cooking initiatives in Nigeria, and thus, the basis for the scenarios are presented in Table 1. According to Table 1, four scenarios are highlighted they are (i) the baseline scenario. The baseline scenario assumes that by 2030, the share of the total clean cooking energy services will be according to the following: Liquefied Petroleum Gas (LPG) at about 54%, Biogas 3%, Briquettes 5%, and Kerosene will be completely phased out by 2060. This assumption is based on data from the Federal Ministry of Environment and clean cooking plans according to the NDCs; (ii) the short-term scenario, the scenario is based on the assumption that about 30% of cooking demand would be met by biogas by 2060 and the remaining demand (i.e., 70%) would be met according to firewood – 20%, charcoal – 15%, briquettes – 10%, sawdust – 5% and LPG will meet about 30% through the simulating period.

Also, a gradual phase-out of the use of kerosene (0% by 2060) is expected by 2060. The scenario implies a uniform reduction rate to remaining biomass fuels and LPG adoption grows significantly (increase to 30%).

The third scenario presented in Table 2 presents an increase in adoption of biogas as an alternative cooking fuel, according to the following assumption. It is expected that biogas would be used to meet about 60% of cooking demand by 2060; the remaining cooking fuel demand shall be met accordingly: charcoal – 20%, briquettes – 10%, Sawdust – 5% and LPG – 5%. Additionally, the key assumption of the third scenario aims to completely phase out the use of kerosene and firewood by 2060, while bioenergy and its fuels other than firewood (charcoal, briquettes, sawdust) account for about 40% to meet the total cooking demand in the SE. The fourth scenario primarily assumes a complete phase-out of firewood, charcoal, briquettes, and sawdust (i.e., all forms of traditional

Table 1 Summary of key features in the scenarios

Scenario	Baseline	Short-term scenario	Medium-term scenario	Long-term scenario
Assumptions	By 2030, share of the total delivered clean cooking energy services will be - Liquefied Petroleum Gas (LPG) 54%, - Biogas 3%, - Briquettes 5%, and - Kerosene will be completely phased out.	- Biogas Contribution: - Set biogas to meet 30% of cooking demand by 2060. - Remaining Cooking Fuels (2060): - Firewood: 20% - Charcoal: 15% - Briquettes: 10% - Sawdust: 5% - LPG: 30% - Key Assumptions: - Gradual phase-out of kerosene (0% by 2060). - Assign a uniform reduction rate to remaining biomass fuels. - LPG adoption grows significantly (increase to 30%).	- Biogas Contribution: - Set biogas to meet 60% of cooking demand by 2060. - Remaining Cooking Fuels (2060): - Charcoal: 20% - Briquettes: 10% - Sawdust: 5% - LPG: 5% - Key Assumptions: - Complete phase-out of kerosene and firewood (0% by 2060). - Biomass fuels other than firewood (charcoal, briquettes, sawdust) account for 40%.	- Biogas Contribution: - Set biogas to meet 90% of cooking demand by 2060. - Remaining Cooking Fuels (2060): - LPG: 10% - Key Assumptions: - Complete phase-out of firewood, charcoal, briquettes, and sawdust (0% by 2060). - LPG serves as a supplementary fuel, growing to meet the remaining 10% demand.

Table 2 Some key assumptions for South-Eastern region Modelling- base year 2023

Item	Assumption				
South-Eastern Region	Abia State	Anambra	Ebonyi	Enugu	Imo
Population	3,100,000	7,250,000	4,500,000	3,470,000	4,820,000
Percent Rural Population (%)	78.7	17.3	11.4	30.8	55.4
Average Household Size	3.28	3.57	5.38	3.26	2.69
Cooking Fuel Share (%)	63.1	60.2	89.2	73.1	88.5
Firewood	0.4	0.8	0	0.6	0.2
Charcoal	2.7	1.8	3.8	1.2	2.1
Briquettes	0.1	0	4.1	0	0.2
Sawdust	1	0.6	0.3	1.1	0.4
LPG	32.5	36.5	2.6	24	8.5
Kerosene	0.2	0.1	0	0	0.1
Biogas					

biomass) by 2060. LPG will serve as a supplementary fuel, growing to meet the remaining 10% demand. It implies biogas will meet about 90% of total cooking energy demand by 2060 and the remaining 10% will be met by LPG.

This research used the baseline scenario as a benchmark for developing three alternative scenarios: the short-term, medium-term, and long-term scenarios. The baseline scenario represents the base case without any energy policy intervention. The alternative scenarios, on the other hand, have had their policy strategies increased in degrees ranging from the short-term scenarios with moderate policies, to a more aggressive medium-term scenario, and finally the long-term scenario, which has a high level of biogas application for meeting heating and cooking needs in the SE. The summary of the key assumption parameters and features of each scenario used in the SE Nigerian LEAP model are presented in Table 2. A base year of 2023 was used to benchmark the simulation procedure of the scenarios. The year 2023 was preferred because the year presents the availability of a complete set of data for the region. Relevant data, such as the rural population percentage and average household size, were obtained from [14]. The cooking fuel share was obtained from the Nigeria data portal through the National Bureau of Statistics (NBS) [38] and are presented in Table 2.

Other data sources included National Bureau of Statistics in Nigeria (NBS) (primary data source), Energy Commission of Nigeria (ECN) (for national policies, some primary data and plans), the World Bank and International Energy Agency (IEA).

3 Results and discussion

The result of the analysis is reported in two major criteria, that is, the energy consumption and the emissions scenarios. The scenarios were further discussed based on the cooking fuels demanded/consumed by the individual state in the SE region.

3.1 Energy consumption scenario

Based on the key features and assumptions about the Southeast region presented in Tables 1 and 2, the energy consumption scenarios were developed according to Fig. 2. Figure 2 shows the Southeast States' cooking energy demand/consumption under the four scenarios from 2023 to 2060. The figure presents the four scenarios of cooking energy demand across the five Southeast states from 2023 to 2060.

According to Table 2, the Imo and Anambra states have the highest proportion of rural population, at around 55.4%, and 78.7% of the total population lives in rural areas.

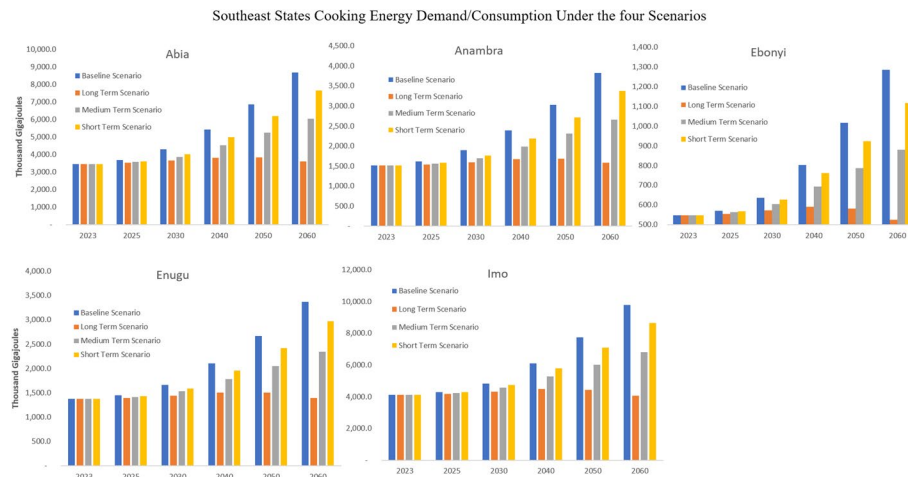


Fig. 2 The Southeast States cooking energy demand/consumption under the four scenarios from 2023 to 2060

This high proportion of rural dwellers has significantly impacted the cooking energy demand presented in Fig. 2. It shows a steady growth in cook energy demand across the five states, however, the states with a high proportion were observed in the states with a higher rural population, predominantly Abia and Imo states. Ebonyi state has the lowest cooking energy demand across all the scenarios. Figure 2 revealed a steady growth in cooking demand from 2025 to 2026 across all scenarios except the base year, 2023.

The cooking demand in the baseline scenario has significantly increased from 2025 by about 150% (from 3458.20k GJ to 8678.90k GJ) in the rural Abia state. However, across all the baselines in the state energy demand, with about 135% in Ebonyi to 153% in Anambra. Similar increase was experienced in the short-term scenario, with around 121% (from 3458.20k GJ in 2023 to 7661.10k GJ in 2060) cooking demand of Abia, meanwhile, other states are with a similar increasing trend in the short-term scenario i.e., a 123%, 104%, 117% and 110% in the Anambra, Ebonyi, Enugu and Imo, respectively.

In the medium-term scenario, with an assumed 60% biogas demand for cooking needs by 2060. The cooking demand grew significantly across all the states at an average of 2800k GJ through the simulated period. Putting this in perspective, states like Anambra, Abia, and Enugu had the highest demand growth in the scenario, with a percentage increase of around 75%, 74% and 71% respectively. Likewise, Ebonyi and Imo states recorded an increase of about 61% and 66% respectively. For the long-term scenario, extremely low cooking demand was recorded across all five states, with Imo and Enugu having the lowest at about 1.2% and 1.9%, respectively. Other states like Anambra, Ebonyi, and Abia states had 4.7%, 4.1% and 3.9%, respectively, recorded a low cooking demand in the Long-term scenario at a demand average of around 1934.5k GJ. The reason for the significantly low cooking energy demand in the long-term scenario could be due to the usage of biogas and the eradication of traditional biomass to meet the cooking energy needs.

3.2 The states' cooking energy demand by fuel based on scenario

The sub-section presents scenario analysis of cooking energy demand by fuel, and the following cooking fuels were considered: kerosene, LPG, firewood, charcoal, biomass and biogas. The scenario based on cooking energy demand by fuels per state was

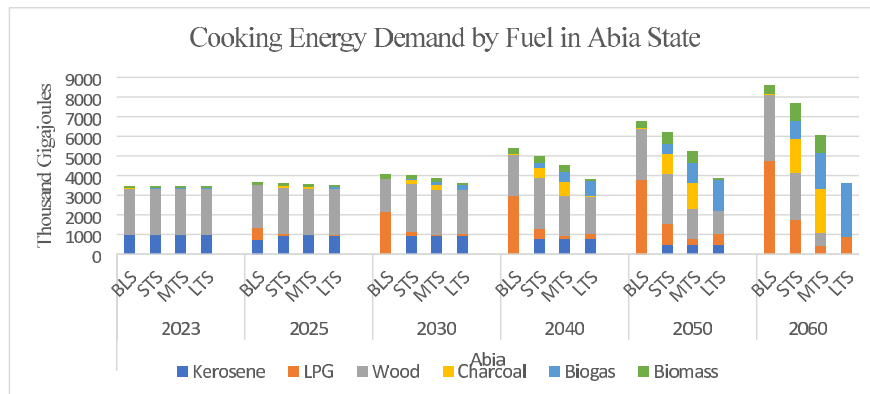


Fig. 3 Abia State cooking energy demand by fuel under the four scenarios

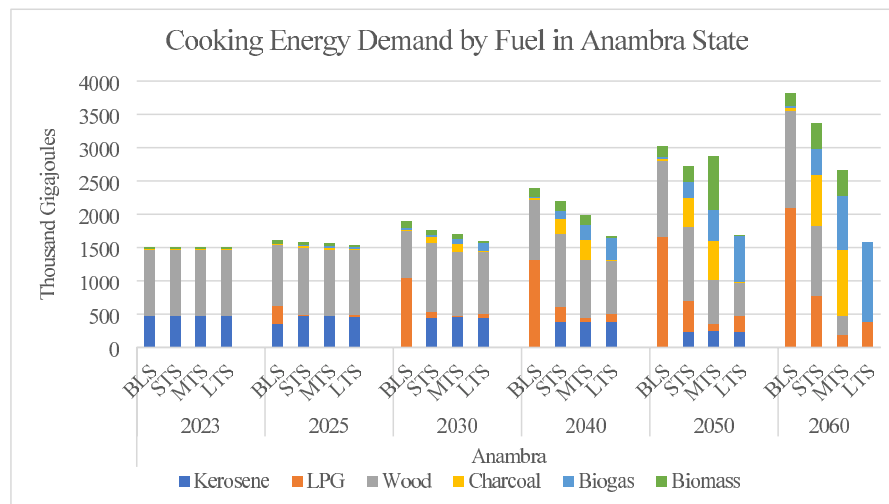


Fig. 4 Anambra State cooking energy demand by fuel under the four scenarios

discussed in Figs. 3, 4, 5, 6 and 7. However, it is important to note that the following abbreviations are employed in the scenario names that is Baseline Scenario as BLS, Short-term Scenario as SLS, Medium-term Scenario as BLS and Long-term Scenario as LTS.

3.2.1 The abia state scenario by fuel analysis

The LEAP analysis for Abia State, shown in Fig. 3, reveals distinct energy demand trajectories under different transition scenarios, reflecting the interplay of fuel-switching dynamics and rural energy access challenges. In the Baseline Scenario, energy demand nearly triples by 2060 (8,679k GJ), driven by significant reliance on traditional biomass (63% firewood, 33% kerosene in 2023) and rural population growth. This trajectory underscores the unsustainable pressure on biomass resources in a state with 78.7% rural households, where inefficient cooking methods dominate. In the Short-term scenario, demand grows moderately (7,661k GJ by 2060), reflecting partial success in kerosene phase-out and LPG adoption (30% share). However, persistent use of firewood (20%), charcoal (15%), and slow biogas rollout (30%) limit efficiency gains. Rural households likely face lingering dependence on biomass despite incremental clean fuel penetration.

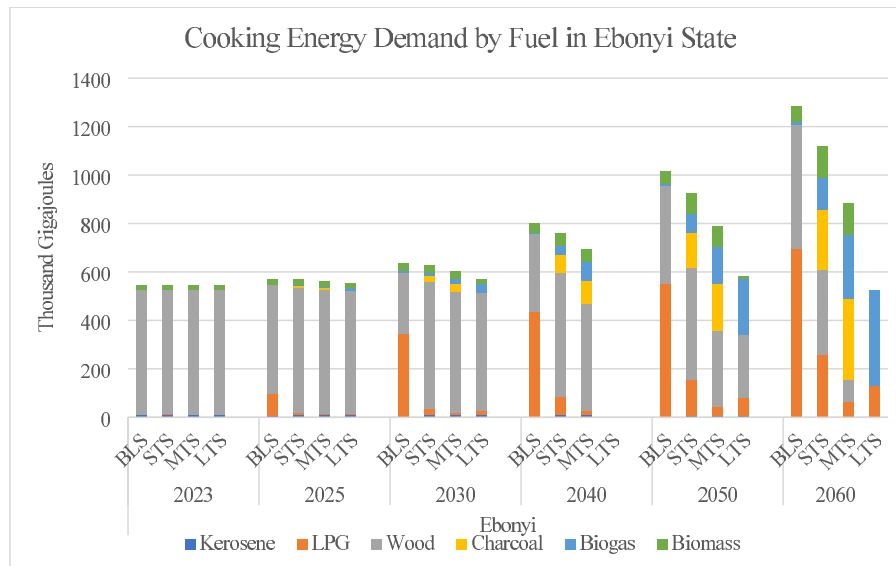


Fig. 5 Ebonyi State cooking energy demand by fuel under the four scenarios

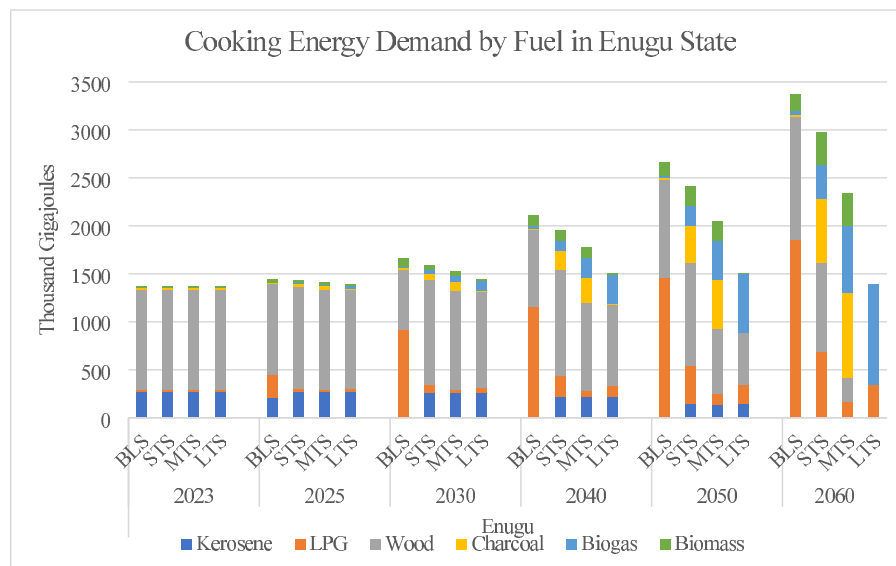


Fig. 6 Enugu State cooking energy demand by fuel under the four scenarios

In the Medium-Term Scenario, it can be observed that demand stabilizes at 6,041k GJ by 2060, aided by aggressive firewood/kerosene elimination and biogas meeting 60% of demand. The residual 40% reliance on charcoal/briquettes (higher energy density than raw biomass) explains why demand remains above the Long-Term Scenario. In the Long-Term Scenario, demand peaks in 2050 (3,837k GJ) before declining to 3,595k GJ by 2060, reflecting near-total biogas adoption (90%) and full phase-out of solid fuels. The post-2050 drop suggests efficiency gains from mature biogas systems and reduced energy losses compared to traditional biomass. However, the initial demand surge (2023–2050) likely mirrors the transitional costs of scaling decentralized biogas plants across rural communities.

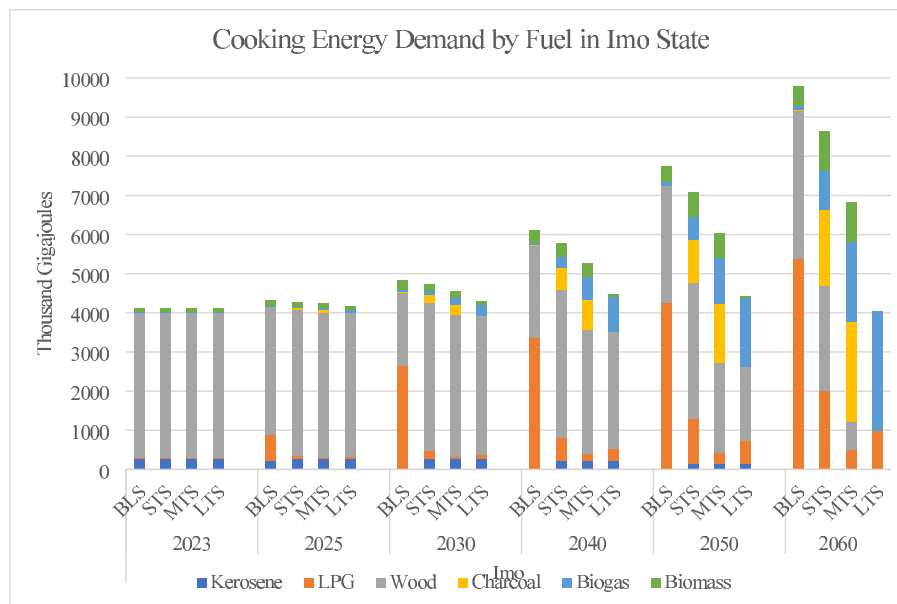


Fig. 7 Imo State cooking energy demand by fuel under the four scenarios

Based on these results, it can be opined that Abia's baseline trajectory is untenable, with biomass-driven demand outpacing sustainable supply. The Long-Term Scenario, while optimal, hinges on unprecedented biogas infrastructure investment and behavioural shifts in rural areas. The Medium-Term Scenario offers a pragmatic balance, reducing firewood dependence without overextending biogas deployment capacity. All pathways highlight the need for parallel investments in stove efficiency and fuel distribution networks to mitigate energy poverty risks during transition.

3.2.2 The Anambra state scenario by fuel analysis

Figure 4 shows the Anambra State scenario analysis based on cooking fuel demand. The baseline scenario in Fig. 4 shows energy demand triples from 1,512.5k GJ in 2023 to 3,826.4k GJ in 2060, driven by a continued reliance on traditional biomass, 60.2% firewood, 36.5% kerosene. This revealed that LPG and traditional biomass are the main drivers of the cooking consumption patterns and evidence of inefficient traditional biomass usage. In the Short-Term Scenario, demand grows to 3,375.2k GJ by 2060 at about 12% lower than baseline. Partial biogas adoption (30%) and LPG growth (30%) slow biomass use but fail to curb overall demand, indicating fragmented progress. Persistent firewood (20% share) and charcoal (15%) sustain inefficiencies.

In the Medium-Term Scenario, peak demand reaches 2,661.2k GJ (2060), 30% below baseline. Aggressive biogas adoption (60%) and firewood phase-out yield significant savings, but residual biomass (charcoal, briquettes) and modest LPG use (5%) limit deeper reductions. The plateau post-2050 suggests diminishing returns without full biomass elimination. For the Long-Term Scenario demand peaks at 1,684.9k GJ (2050) before declining to 1,583.7k GJ (2060) – 59% below baseline. Near-total biogas reliance (90%) and elimination of all biomass fuels drive this reversal, underscoring the transformative potential of waste-to-energy systems. The post-2050 dip signals efficiency gains from mature biogas infrastructure, but achieving this requires upfront investment in biogas grids and waste management.

3.2.3 The Ebonyi state scenario by fuel analysis

The LEAP analysis for Ebonyi State, shown in Fig. 5 reveals divergent energy demand trajectories under varying transition strategies, with critical implications for policy prioritization in a region dominated by biomass dependence at about 89.2% of firewood use. For the Baseline Scenario, demand surges 136%, from 546.4k GJ (2023) to 1,285.9k GJ (2060). This reflects unmanaged reliance on firewood and kerosene, exacerbated by population growth and stagnant fuel efficiency. The steep post-2030 rise signals systemic inertia, with traditional biomass locking in high energy intensity. In the Short-Term Scenario, cooking demand grows to 1,117.2k GJ by 2060 at about 13% lower than baseline but still doubling 2023 levels. Partial biogas adoption by 30% and LPG growth slows firewood use but fails to disrupt traditional biomass dominance, particularly in sparse rural clusters of around 11.4% rural population.

Also, residual inefficiencies persist due to charcoal's share of about 15% share. The Medium-Term Scenario sees demand peaking at 880.9k GJ in 2060, which is about 32% below baseline. Firewood phase-out and 60% biogas adoption drive progress, but residual charcoal/briquettes (40% combined) sustain elevated demand. The plateau post-2050 suggests diminishing returns without full biomass elimination. In the Long-Term Scenario, demand peaks at 581.1k GJ by 2050 before declining to 524.2k GJ by 2060, which is about 4% below 2023 levels. Near-total biogas reliance (90%) and elimination of all biomass fuels enable this reversal, highlighting the transformative potential of waste-to-energy systems. The post-2050 dip reflects efficiency gains from mature biogas infrastructure.

3.2.4 The Enugu state scenario by fuel analysis

Figure 6 reveals the Enugu state cooking energy demand by fuel type under the four scenarios. According to Fig. 6, the Baseline Scenario of cooking energy demand triples from 1,369.3k GJ in the base year (2023) to 3,369.3k GJ in 2060. This reflects unmanaged dependence on firewood and kerosene (24% initial share), with urbanization driving inefficient LPG adoption and persistent biomass use. The steep post-2030 rise signals systemic inertia in fuel transitions. In the Short-Term Scenario, demand grows to 2,973.7k GJ by 2060—12% below baseline but still doubling 2023 levels. Partial biogas adoption (30%) and LPG growth (30%) slow firewood use, but residual biomass (20% firewood, 15% charcoal) sustains inefficiencies. Peri-urban and rural clusters resist full transition due to affordability gaps. For the Medium-Term Scenario, demand peaks at 2,344.7k GJ (2060), 30% below baseline. Firewood phase-out and 60% biogas adoption drive progress, but residual charcoal/briquettes (40% combined) and modest LPG (5%) limit gains. Post-2050 plateau suggests diminishing returns without full biomass bans. For the Long-Term Scenario, demand peaks at 1,503.2k GJ (2050) before declining to 1,395.4k GJ (2060) – 2% below 2023 levels. Near-total biogas reliance (90%) and elimination of all biomass fuels enable this reversal, showcasing the transformative potential of waste-to-energy systems. The post-2050 dip reflects efficiency gains from mature biogas infrastructure.

The Imo State Scenario by Fuel Analysis.

Figure 7 presents the Imo state cooking fuel demand under the four scenarios. It revealed that the main fuel types in the base year, 2023, are kerosene and traditional biomass, with traces of LPG and charcoal. The BLS revealed a soaring of energy demand

more than from 4,107.1k GJ in the 2023 base year to 9,788.1k GJ in 2060. This reflects unmanaged reliance on firewood and kerosene, exacerbated by population growth and stagnant fuel efficiency. The steep rise signals systemic inertia, with traditional biomass locking in high energy intensity and inefficiency. In the STS, cooking demand grows to 8,641.6k GJ by 2060—12% below baseline but still doubling 2023 levels. Partial biogas adoption (30%) and LPG growth (30%) slow firewood use but fail to disrupt biomass dominance.

According to Fig. 7, the STS revealed residual inefficiencies persist due to charcoal substitution of about 15% and fragmented rural access to clean fuels. In the MTS, cooking demand peaks at 6,813.6k GJ (2060), 30% below baseline. Firewood phase-out and 60% biogas adoption drive progress, but residual charcoal/briquettes (40% combined) sustain elevated demand. The post-2050 growth suggests diminishing returns without full biomass elimination or infrastructure scalability. In the LTS, cooking demand peaks at 4,433.6k GJ in 2050 before declining to 4,054.9k GJ in 2060 at 1.3% below 2023 levels. Near-total biogas reliance (90%) and elimination of all biomass fuels enable this reversal, showcasing the transformative potential of waste-to-energy systems. The post-2050 dip reflects efficiency gains from mature biogas infrastructure and reduced energy waste.

The STS and MTS across all states revealed a moderately low cooking energy demand compared to the BLS. This is due to the growth in the use of more efficient cooking fuel, such as biogas and LPG in the cooking fuel mix. It therefore implies that traditional biomass and kerosene should be reduced and eradicated to achieve a more sustainable clean cooking sector in the SE, Nigeria. Further, the LTS, according to all the states under consideration in the modelling period, revealed that cooking energy demand would be reduced tremendously with the use of more efficient cooking fuels such as biogas and LPG. Indeed, these cooking fuels are drivers of the needed energy transition in the heating and cooking sector in the Southeastern region and Nigeria as a whole. However, there is a need for further studies and analysis to validate this claim for the Nigerian clean cooking transition. However, other relevant factors for the Nigerian clean cooking transition were modelled in [42].

3.3 Emission scenarios

Figure 8 shows emission scenarios across the five states in the Southeast of Nigeria. Results from the LEAP analysis revealed a clear trajectory of emissions under different scenarios for all States in Southeast Nigeria, from 2023 to 2060. In the Baseline Scenario, emissions rise steadily, reflecting a business-as-usual approach where traditional biomass fuels like firewood and kerosene remain dominant.

Abia and Imo states showed the highest trend in carbon emissions as a result of inefficient cooking fuel and technology, and also a significantly high proportion of the rural population. When compared with the two states earlier stated, Anambra, Enugu, and Ebonyi have lower emission trends, in that order. Across all five states, the Baseline scenario shows a very high emission value compared to other scenarios. For instance, by 2060, emissions reached an average of 384.8 thousand metric tonnes of CO₂ equivalent, highlighting the environmental cost of inaction. In contrast, the Short-Term Scenario shows a slower but still significant increase in emissions, peaking at 224k MtCO₂e by 2060 – a reduction of approximately 42% compared to the baseline. The Medium-Term Scenario further curbs emissions, reaching 178k MtCO₂e by 2060, driven by a

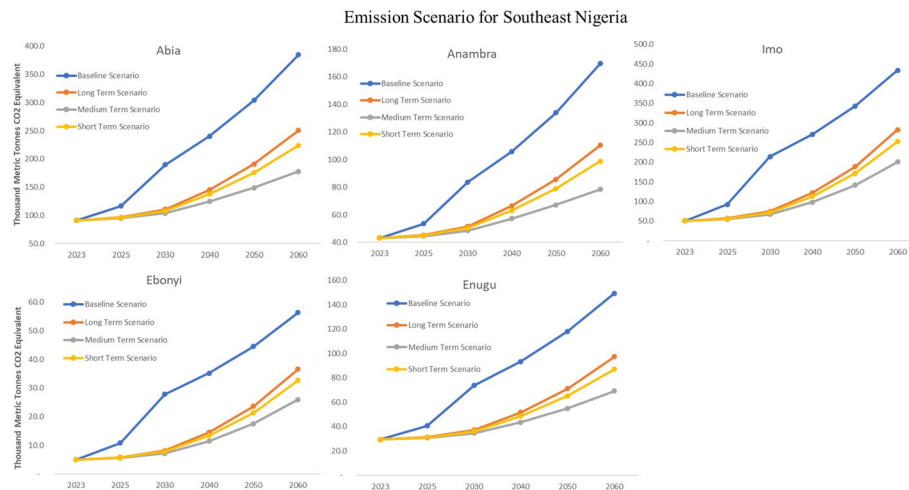


Fig. 8 Emissions scenario for Five Southeast States in Nigeria

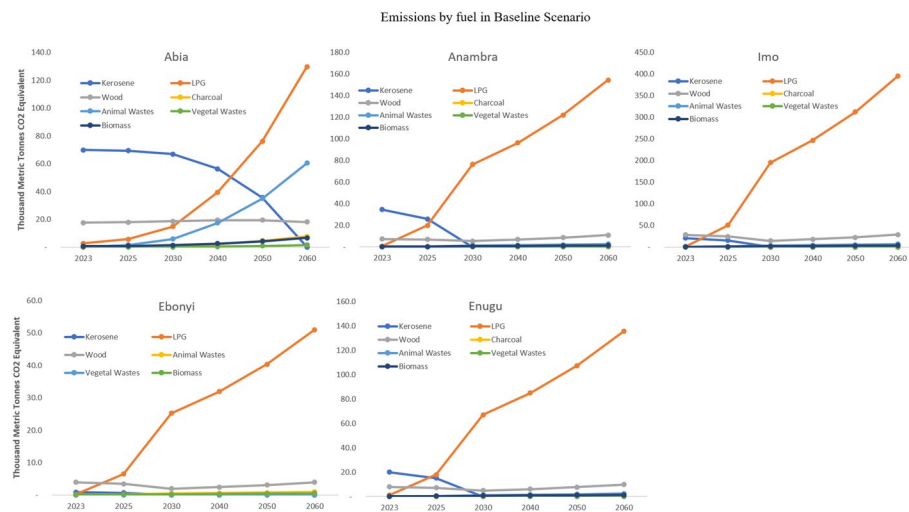


Fig. 9 Southeast States emissions by fuel in Baseline Scenario

more aggressive adoption of biogas (60% of cooking demand) and a complete phase-out of firewood and kerosene. Emissions reductions here are roughly 54% lower than the baseline. The Long-Term Scenario shows emissions reaching 250.6k MtCO₂e by 2060 – a reduction of approximately 35% compared to the baseline. This scenario envisions a near-complete transition to biogas (90% of cooking demand) and LPG as a supplementary fuel, alongside the total elimination of firewood, charcoal, briquettes, and sawdust. While emissions are higher than in the Medium-Term Scenario, this reflects the challenges of scaling biogas infrastructure and maintaining LPG as a backup. Meanwhile, the proportion of emission decrease is similar between the baseline-term, short-term, medium-term and long-term scenarios for all states (Fig. 9).

Additionally, the Anambra baseline Scenario in 2060, with around 169.5k MtCO₂e, emissions rise steadily, reflecting unmanaged reliance on firewood at 60.2% of cooking fuel and kerosene (36.5%). By 2060, emissions nearly quadruple from 2023 levels, underscoring the environmental cost of stagnant fuel policies and urban-rural disparities at 17.3% rural population. In the Short-Term Scenario in Fig. 10, about 98.7k MtCO₂e,

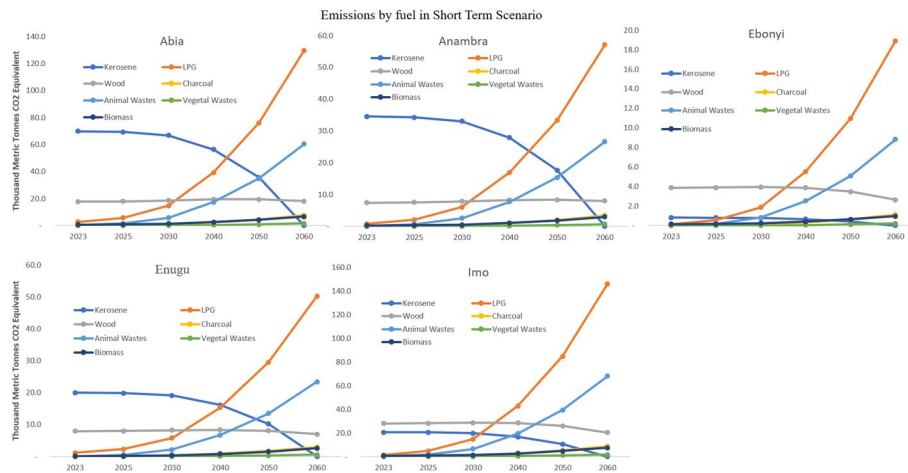


Fig. 10 Southeast States emissions by fuel in the short-term Scenario

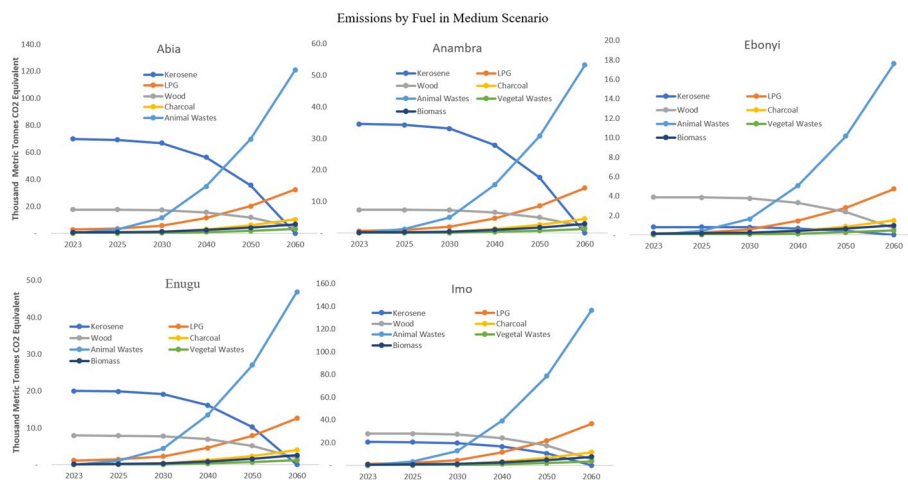


Fig. 11 Southeast States emissions by fuel in the medium-term Scenario

kerosene is phased out by 2060, but lingering firewood use and rising LPG adoption drive emissions. While a 42% reduction from the baseline is achieved, persistent biomass dependence (charcoal, sawdust) and moderate biogas adoption (30%) limit progress. Emissions grow post-2040, highlighting systemic inertia in rural fuel transitions. For the Medium-Term Scenario in Fig. 11, about 78.4k MtCO₂e of emission was obtained due to an aggressive biogas adoption (i.e., 60% by 2060) and firewood phase-out yields the steepest reduction at about 54% below baseline. Emissions plateau after 2050, but residual charcoal use (20% share) and biomass were linked to emissions in that year (e.g., vegetal wastes), revealing persistent challenges in displacing traditional fuels. Looking at the Long-Term Scenario as shown in Fig. 12 (i.e., in 2060, around 110.4k MtCO₂e emissions), despite biogas meeting 90% of demand, emissions exceed the Medium-Term Scenario (shown in Fig. 11). This paradox likely stems from methane leakage in biogas systems or increased processing of animal/vegetal wastes, offsetting gains from phasing out firewood and charcoal. LPG's supplementary role at 10% adds modest emissions but ensures energy security. These results provide some key insights: (i) Biogas Trade-Offs - while critical for decarbonization, biogas scalability in Anambra – a state with

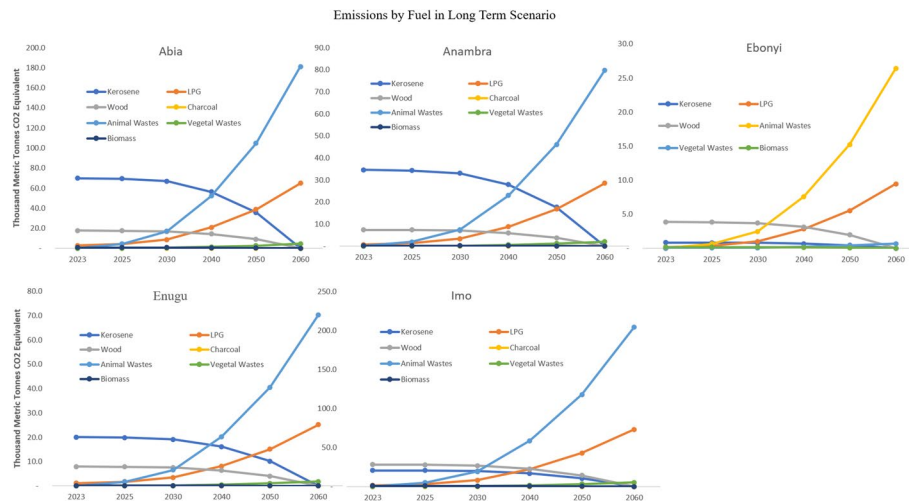


Fig. 12 Southeast States emissions by fuel in the long-term Scenario

low rural population but high kerosene use – requires addressing methane leakage and waste-processing emissions; (ii) The Medium-Term Scenario balances feasibility and impact, but achieving Long-Term goals demands stricter enforcement of biomass bans and cleaner biogas tech; and (iii) Kerosene phase-out risks affordability shocks for low-income households; targeted subsidies for biogas/LPG are essential.

In the Ebonyi Baseline Scenario emissions shown in Fig. 8 emissions skyrocket 11-fold, from 5.0k MtCO₂e in 2023 to 56.3k MtCO₂e in 2060. This mirrors unmanaged reliance on firewood (89.2% of cooking fuel) and kerosene (2.6%), exacerbated by limited clean fuel access in rural areas. The steep post-2030 rise signals entrenched biomass dependence and a lack of policy intervention. For the Short-Term Scenario, emissions reach 32.7k MtCO₂e by 2060—42% lower than baseline but still 6times 2023 levels. Gradual kerosene phase-out and 30% biogas adoption curb growth, but residual firewood (20%) and rising LPG (30%) sustain emissions. Rural inertia and fragmented clean fuel distribution stall deeper cuts. In the Medium-Term Scenario, aggressive biogas adoption (60%) halves emissions to 26.0k MtCO₂e (2060). Firewood phase-out and LPG reduction (5%) drive progress, but residual charcoal/briquettes (40% combined) reveal substitution challenges. Emissions plateau post-2050, suggesting diminishing returns without full biomass elimination.

While in the Long-Term Scenario emissions rise to 36.5k MtCO₂e by 2060 – paradoxically higher than Medium-Term. While biogas meets 90% demand, methane leakage from intensified waste processing likely offsets gains. Full biomass phase-out eliminates firewood/charcoal, but LPG's supplementary role (10%) adds emissions.

These results illuminate the following key insights: (i) Ebonyi's low rural population (11.4%) paradoxically complicates transitions – concentrated biomass use in sparse communities resists centralized interventions; (ii) Medium-Term's balance of feasibility and impact is optimal; (iii) Short/Medium-Term's lingering charcoal/briquettes highlight systemic gaps in displacing traditional fuels without economic incentives; (iv.) Kerosene phase-out risks excluding remote households; decentralized biogas microgrids are critical for inclusivity. Furthermore, Ebonyi State's results reveal that even modest rural populations can drive high emissions without targeted policies. Thus, Ebonyi State showed the least emissions scenarios across all four scenario trends. This low emission scenario

is a result of a low rural area proportion of 11.4% of the entire population (as shown in Table 2).

The Enugu State reveals critical disparities in emission trajectories under varying transition strategies, shaped by its moderate rural population (30.8%) and heavy reliance on firewood (73.1% of cooking fuel). The Baseline Scenario shows emissions surge 5-fold, from 29.3k MtCO₂e (2023) to 149.3k MtCO₂e (2060). This reflects unmanaged dependence on firewood and kerosene (24% share), with urbanization driving LPG adoption inefficiencies. The steep post-2030 rise signals policy inertia and biomass lock-in. The Short-Term Scenario shows that emissions reach 86.9k MtCO₂e by 2060—42% lower than baseline but still 3 times 2023 levels. Gradual kerosene phase-out and 30% biogas adoption slow growth, but persistent firewood (20% share) and rising LPG (30%) sustain emissions. For the Medium-Term Scenario, aggressive biogas adoption (60%) halves emissions to 69.1k MtCO₂e (2060). Firewood phase-out and LPG reduction (5%) drive gains, but residual charcoal/briquettes (40% combined) reveal substitution gaps. Emissions plateau post-2050, signalling diminishing returns without full biomass bans. For the Long-Term Scenario, emissions rise to 97.3k MtCO₂e by 2060 – exceeding Medium-Term despite 90% biogas use. Likely causes: methane leakage from intensified waste processing and LPG's supplementary role (10%). Full biomass phase-out eliminates firewood/charcoal but underscores biogas system inefficiencies [23].

In Fig. 8, under the Baseline Scenario (business-as-usual trajectory), the Imo State emissions rise sharply from 50.6 thousand metric tonnes CO₂e in 2023 to 434.1 thousand metric tonnes by 2060, reflecting unchecked reliance on firewood (88.5% current share) and kerosene. This 8.5-fold increase aligns with population growth and persistent biomass use, which dominates rural cooking practices. In the Short-Term Scenario (30% biogas by 2060), emissions grow to 252.7 thousand metric tonnes CO₂e by 2060—42% lower than Baseline. Early kerosene phase-out and moderate biogas adoption curb emissions but fail to offset rising demand from residual biomass (charcoal, sawdust). In the Medium-Term Scenario, emissions stabilise at 200.7 thousand metric tonnes by 2060, driven by stricter firewood phase-out and expanded biogas infrastructure. This pathway cuts Baseline emissions by 54%, highlighting the outsized role of eliminating firewood (a high-emission, low-efficiency fuel). In the Long-Term Scenario (90% biogas by 2060), emissions peak at 282.7 thousand metric tonnes by 2060, still 35% below Baseline. Surprisingly, this pathway underperforms the Medium-Term Scenario, likely due to delayed LPG phase-out (retained at 10%) and slower biogas scale-up relative to population growth.

3.4 Emissions by fuel based on scenarios

The baseline scenario emissions for Abia State (according to Fig. 9) reveal critical insights into the trajectory of household cooking fuel use and its environmental impact under current assumptions. By 2030, kerosene emissions will drop to zero, aligning with its phased elimination. However, this is offset by a sharp rise in LPG emissions, which grow from 2.7 by 2023 to 350.0 thousand metric tonnes CO₂e by 2060. This reflects a heavy reliance on LPG as a transitional fuel, but its growing emissions signal inefficiencies in adoption (e.g., incomplete combustion, leakage) or lack of complementary clean energy integration. Wood remains a persistent contributor, with emissions rising from 17.7 in 2023 to 25.1 by 2060, likely driven by rural population reliance (78.7%) and slow

displacement of traditional biomass. Minor contributors like charcoal, animal/vegetal wastes, and general biomass show negligible growth, underscoring their limited role in the baseline trajectory. The results highlight a paradox: while kerosene phase-out reduces emissions initially, unchecked LPG dependency and stagnant biomass use create a net emission increase of 283% by 2060. This underscores the inadequacy of a baseline strategy focused solely on fuel substitution without addressing efficiency, renewable integration (e.g., biogas), or rural energy access. For meaningful decarbonization,

Abia State must accelerate transitions beyond LPG to truly low-carbon solutions while tackling structural drivers of biomass dependence.

Fuel used in meeting cooking and heating demand in the region; the emissions effect of cooking fuel is illustrated in Figures 9 through 12. Figures 9 and 10 show the emissions by fuel in the Short-term scenario. In this scenario, LPG as a cooking fuel source shows significant emissions across all the States; however, at different emission rates, it is clear that all states had the same increasing trend. This could be due to the consistent use of LPG (at 30%), an alternative cooking fuel. Also, animal waste increased alongside the LPG; however, at different trajectories across all the States, this is likely due to reliance on Biogas, as the main constituent of biogas is animal waste. In the Baseline and Short-term scenarios, LPG is the major cooking fuel across all the states, thus the main reason for emissions from LPG.

The emissions by cooking fuel in the medium and long-term are illustrated in Figs. 11 and 12, respectively. In the medium-term scenario represented in Fig. 11, emissions due to animal waste are prominent across all the States. The figure shows a tremendous growth trend as a result of high use of biogas (about 60%) to meet cooking demand. However, emissions from traditional biomass, including charcoal and fuelwood, are not the major carbon and pollutant emitters, as they have lower emissions compared to LPG. Despite emission reduction in kerosene and an eventual eradication by 2060, there is a gain of emissions through other sources such as biogas, which is expected to emit less. This is largely due to overdependence on animal waste as fuel that produces the biogas.

Figure 12 illustrates emissions by fuel in the long-term scenario, which has similar trend characteristics to those represented in Fig. 11. Thus, a significant increasing trend in emissions from animal waste followed by emissions from LPG, with a reducing emission trend from traditional biomass and kerosene. This is largely due to the over-reliance on biogas (90%) and LPG (10%) while eradicating other cooking fuel sources [27].

4 Discussion of limitations

The study has been limited by several factors, which included: (i) data quality and availability, the numerical data used in the study are estimates and a few from empirical analysis, as there were limited regional data, especially for the SE region of Nigeria. Also, the non-availability of regional data has had significant consequences on the quality of data, hence the outcome of the study; (ii) technical and institutional barriers to bioenergy deployment; technology development has prevented and slowed the rate of research in sustainable cooking in the region as there are a few or no institutions coordination and regulation clean cooking, as such efficient understanding and penetration of the subject matter is limited reducing the quality of data and stakeholder engagement in the study. More so, there is limited technological know-how in the sector. Iii) possible oversimplification of behavioural or socio-economic dynamics in the modelling process:

there are possibilities of overstated or oversimplified results due to data quality and weak institutions that could have ensured results validation and model calibration. Also, consumer and household behaviour, including socio-economic dynamics of the region, has impacted the modelling processes and subsequently their outcome. Additionally, the current study is strongly based on assumptions that are based on published reports from ministries and agencies in Nigeria; however, there are no regional ministries that manage data, reports and other valuable information on clean cooking in Nigeria. Hence, the national reports have been downscaled based on some socioeconomic and technology indicators to fit the regional realities, thus introducing significant anomalies and under-/overstated results. These limitations are particularly relevant given the rural context of the study.

5 Conclusion and policy implications

This study examined the potential pathways for energy demands and CO₂ emission abatement through scenario analysis of cooking fuel transitions and bioenergy utilization in rural Southeast Nigeria. The findings highlight that shifting from traditional biomass fuels such as firewood and charcoal to cleaner alternatives, including liquefied petroleum gas (LPG), biogas, and improved biomass cookstoves, can substantially enhance energy efficiency while potentially reducing household CO₂ emissions. Scenario projections suggest that strategic interventions promoting affordable, accessible, and sustainable cooking fuel options, especially biogas and LPG, can play a critical role in ensuring energy efficiency and significantly cutting cooking energy demand by over 50% in the long run at the community level, meeting rural energy demands. In the study, all transition pathways highlight the need for parallel investments in stove efficiency and fuel distribution networks to mitigate energy poverty risks during transition. Moreover, the assessment underscores the impact of the overutilized potential of locally available bioenergy resources, especially animal waste, to generate biogas, which could potentially lead to more emissions than expected. The over-dependence on biogas could trigger unexpected impact on emissions and lead to what is referred to as a paradox, in the sense that instead of a heavy use of biogas to meet the cooking and heating demand in rural SE, it could otherwise be a source of major emissions in the community. This could be based on the fact that most rural areas in Nigeria lack little or no technology and infrastructure to effectively and efficiently convert bioenergy into sustainable biogas. And this could jeopardise the circular economy programmes that reduce environmental degradation and make achieving sustainable development goals difficult. Overall, this research contributes valuable insights into sustainable energy planning for rural communities, reinforcing the importance of the overexploitation of bioenergy resources in the drive toward low-carbon rural development. The study concludes that the long-term scenario is the captured relevant sustainable outcome of a clean cooking paradigm in SE Nigeria. The scenario ensured reduced cooking demand across all states as a result of highly efficient and clean fuel. Also, low emissions were identified in the long-term scenario. Hence, the long-term scenario is the most appropriate for sustainable socio-economic or infrastructural clean cooking conditions.

Based on the findings of this study, the following recommendations are gleaned from the study: (i) promotion of clean and a mix of sustainable cooking fuels: relevant policies should prioritize the adoption of cleaner bioenergy alternatives such as biogas, briquette

and LPG as overdependency one might not be sustainable, improved biomass stoves, and liquefied petroleum gas (LPG); (ii) strengthening Rural Energy Infrastructure. Infrastructure development plays a crucial role in developing a clean cooking paradigm that would enhance low-carbon development, especially in rural communities of developing countries; (iii) public awareness and capacity building. How to use and deploy the technology is as crucial as the technology itself, so there is need for training and training of users of these clean cook technologies and (iv) further research and development - Future work should further explore the socio-cultural dynamics of fuel adoption, alongside quantitative modelling of long-term emission trajectories under varying policy frameworks and policies focusing on rural energy access, financial incentives for cleaner cookstove adoption, and investments in biogas infrastructure are essential for enabling this transition.

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Author contributions

N.N.C. conceptualized the project, designed the methodology, conducted the literature review and data analysis, performed data collection and analysis, and further managed communication with the journal, M.S. helped in developing the methodology and supervised the study. O.E. assisted in data collection, conclusion and developing recommendations. I.M.M. developed the research objectives, reviewed and edited the manuscript. D.N.S.A. validated the research objective, reviewed and edited the manuscript.

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Data availability

Data will be available upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All Authors have agreed to publish this paper.

Competing interests

The authors declare no competing interests.

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